Ecosystem modeling: Individual-based model

II. Adding physiological variables

Jang-Geun Choi

Center for Ocean Engineering
University of New Hampshire

May, 2026

Recap

One of the simplest (but powerful) individual-based model is particle tracking algorithm[1]:

(Simple) particle tracking algorithm

$$\frac{dX_i}{dt} = U_i \tag{1a}$$

$$\frac{dY_i}{dt} = V_i \tag{1b}$$

where

X_i and Y_i: x- and y-position of i-th individual (particle)
U_i and V_i: each velocity component at the position (X_i, Y_i).

[1] Note that there are no biological terms yet.

Adding an individual-level variable

One of the simplest (but powerful) individual-based model is particle tracking algorithm:

(Simple) particle tracking algorithm with IBM

$$\frac{dX_i}{dt} = U_i \tag{2a}$$

$$\frac{dY_i}{dt} = V_i \tag{2b}$$

$$\boxed{\frac{dQ_i}{dt} = \sum \underbrace{SS}_{\text{Source \& Sink}}} \tag{2c}$$

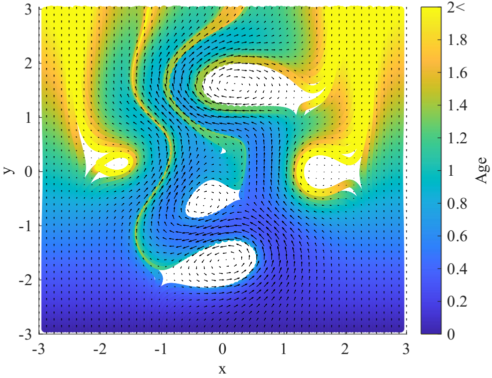
where

X_i and Y_i: x- and y-position of i-th individual (particle)
U_i and V_i: each velocity component at the position (X_i, Y_i).
Q_i: Biological/physiological quantity of interest

Example: age and maturity

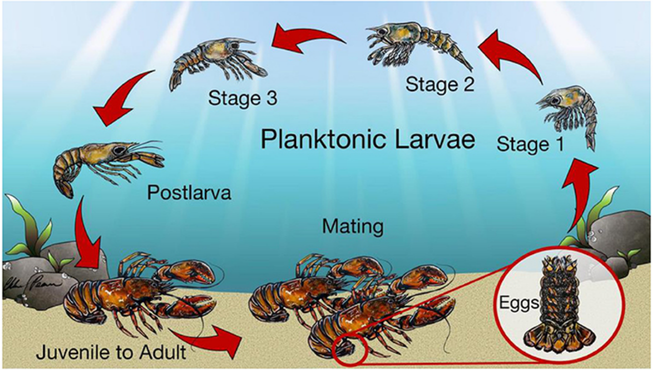
Age of particle (individual) can be modeled by

$$\boxed{\frac{dA_i}{dt} = 1} \rightarrow A_i = t + A_i(0) \tag{3}$$



Example: age and maturity

Life cycle of lobsters: three planktonic stages (I, II, and III), one active swimming stage (postlarva), and one benthic stage (adult).



Source: Watson et al. (2023)

Example: age and maturity

We may model physiological age (maturity) that depends on environmental variables (e.g., temperature).

For example, MacKenzie (1988) suggested formulation for duration of each life stage of lobster larvae $D(T)$, depending on temperature T. This was used by IBM developed by Xue et al. (2008):

$$\frac{dM_i}{dt} = \frac{1}{D_i} \rightarrow M_i = \int_0^t \frac{1}{D_i}\, dt \tag{4}$$

where M_i indicates the maturity. $M_i(t) - M_i(0) = 1$ represents that the individual is fully matured (and moves to the next life stage).

Example: weight

$$\frac{dW_i}{dt} = U_i(F, \boxed{W_i}, ...)W_i \tag{5}$$

where W_i is weight of an individual and U_i indicates "net" uptake rate depending on environmental factors (e.g., amount of food F) and condition of the individual (e.g., W_i).

Usually, U_i is defined as superposition of various physiological processes rate.

There is sophisticated modeling approach: bioenergetics. See Megrey et al. (2007) for an example formulating U_i.

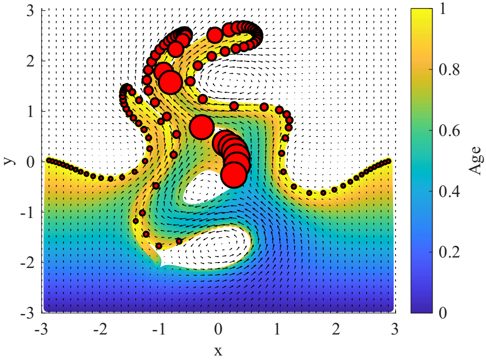
Lab 1

Let us consider a simple bioenergetics model[2]:

$$\frac{dW_i}{dt} = \left(\overbrace{C_{max}\frac{F_i}{F_i + K_F}}^{\text{(Food) consumption}} - \underbrace{a{W_i}^{-b}}_{\text{Respiration}} \right) W_i \qquad (6)$$

with simple analytical food concentration map F:

$$F = e^{-(x^2+y^2)} \qquad (7)$$



References I

MacKenzie, Brian R (1988). “Assessment of temperature effects on interrelationships between stage durations, mortality, and growth in laboratory-reared Homarus americanus Milne Edwards larvae”. In: *Journal of Experimental Marine Biology and Ecology* 116.1, pp. 87–98.

Megrey, Bernard A et al. (2007). “A bioenergetics-based population dynamics model of Pacific herring (Clupea harengus pallasi) coupled to a lower trophic level nutrient–phytoplankton–zooplankton model: description, calibration, and sensitivity analysis”. In: *Ecological modelling* 202.1-2, pp. 144–164.

Watson, WH et al. (2023). “How Might Climate Change Affect American Lobsters”. In: *Front. Young Minds* 11, p. 1031267.

References II

Xue, Huijie et al. (2008). "Connectivity of lobster populations in the coastal Gulf of Maine: Part I: Circulation and larval transport potential". In: *Ecological modelling* 210.1-2, pp. 193–211.